$x = \rho \operatorname{sen}\phi \cos\theta$

$y = \rho \operatorname{sen}\phi \operatorname{sen}\theta$

$z = \rho \cos\theta$

Jacob $= \rho^2 \operatorname{sen}\phi$

$x^2+y^2+z^2=\rho^2$

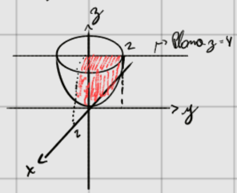
↳ coord esféricos

$x = r\cos\theta$ $\quad x^2+y^2=r$

$y = r\operatorname{sen}\theta$ $\quad dV = r\,dr\,d\theta\,dz$

$z = z$

① $\iiint_S x\,dx\,dy\,dz$ limitado por $z = x^2+y^2$; $z = 2$ no primeiro octante



$z = x^2+y^2 = r^2 \to r^2 = 2 \to r = \sqrt{2}$

$$\begin{cases} r^2 \le z \le 2 \\ 0 \le \theta \le \frac{\pi}{2} \\ 0 \le r \le \sqrt{2} \end{cases}$$

$$\int_0^{\frac{\pi}{2}}\int_0^{\sqrt{2}}\int_{r^2}^{2} r\cos\theta \cdot r\,dz\,dr\,d\theta = \int_0^{\frac{\pi}{2}}\int_0^{\sqrt{2}} r^2\cos\theta\,(2-r^2)\,dr\,d\theta = \int_0^{\frac{\pi}{2}}\int_0^{\sqrt{2}} 2r^2\cos\theta - r^4\cos\theta\,dr\,d\theta = \text{I}$$

$$\text{I} \quad \int_0^{\frac{\pi}{2}}\int_0^{\sqrt{2}} 2r^2\cos\theta - r^4\cos\theta\,dr\,d\theta = \int_0^{\frac{\pi}{2}} \cos\theta\left(\frac{8\sqrt{2}}{15}\right)d\theta = \frac{8\sqrt{2}}{15}\cdot \operatorname{sen}\theta\Big|_0^{\frac{\pi}{2}} = \frac{8\sqrt{2}}{15} \;//$$

$2r^2 - r^4$

$20\sqrt{2} - 12\sqrt{2}$

$$\frac{2r^3}{3}\Big|_0^{\sqrt{2}} - \frac{r^5}{5}\Big|_0^{\sqrt{2}} = \frac{2\sqrt{2}^3}{3} - \frac{\sqrt{2}^5}{5} = \frac{10\sqrt{2}^3 - 3\sqrt{2}^5}{15} = \frac{10\cdot 2\sqrt{2} - 3\cdot 4\sqrt{2}}{15} = \frac{8\sqrt{2}}{15}$$

② $\iiint \sqrt{x^2+y^2}$ limitado por $x^2+y^2=1$; $z = 1-x^2-y^2$; $z = 4$

$\boxed{z = 1-r^2}$

$x^2+y^2 = r^2 \to r = 1$

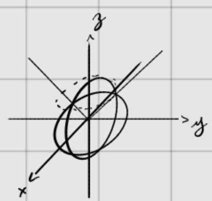
$$\begin{cases} 0 \le \theta \le 2\pi \\ 0 \le r \le 1 \\ 1-r^2 \le z \le 4 \end{cases}$$

Vamos ter a integral $\displaystyle\int_0^{2\pi}\int_0^{1}\int_{1-r^2}^{4} r\cdot r\,dz\,dr\,d\theta = \int_0^{2\pi}\int_0^{1} 3r^2 + r^4\,dr\,d\theta = \int_0^{2\pi}\frac{6}{5}\,d\theta = \frac{12\pi}{5}$ //

z entra pelo parábolo e sai pelo plano, como fizemos a alteração $r = x^2+y^2$, temos que nos atentar aos limites

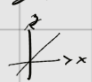
③ $\iiint z\,dx\,dy\,dz$ limitado por $z = \sqrt{x^2+y^2}$; $z = \sqrt{8-x^2-y^2}$

→ CONE ($z = r$) → MEIA ESFERA ($z = \sqrt{8-r^2}$) → $z^2 = 8-x^2-y^2 \to x^2+y^2+z^2 = 8$

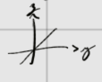


xy: $x^2+y^2=0$, $x=y=0$

xz: $z = x$

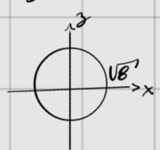


yz: $z = y$

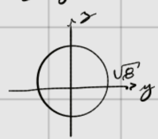


$x^2+y^2-z^2=0$

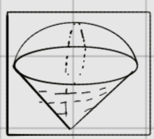
xz: $z^2 = 8-x^2$; $z^2+x^2 = 8$



yz: $z^2 = 8-y^2$; $z^2+y^2 = 8$



Vamos ter uma meia esfera positiva com um cone gerando nossa região ↳



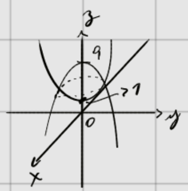
Com isso vamos ter as variações limitadas por:

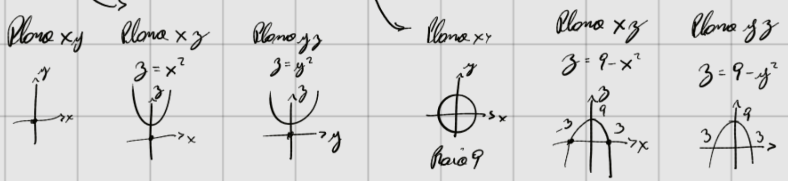
$$\begin{cases} r \le z \le \sqrt{8-r^2} \\ 0 \le r \le \sqrt{8} \\ 0 \le \theta \le 2\pi \end{cases} \quad dz\,dr\,d\theta$$

$x^2+y^2+1 = 9-x^2-y^2$

$2x^2+2y^2 = 8 \to x^2+y^2 = 4 \to \boxed{r = 2}$

④ Determine o volume do sólido delimitado por $z = \overset{r^2}{x^2+y^2}+1$; $z = \overset{9-r^2}{9-x^2-y^2}$





$$\begin{cases} 1+r^2 \le z \le 9-r^2 \\ 0 \le \theta \le 2\pi \\ 0 \le r \le 2 \end{cases} \quad dz\,dr\,d\theta$$

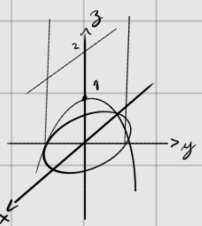
$$\int_0^{2\pi}\int_0^{2}\int_{r^2+1}^{9-r^2} r\,dz\,dr\,d\theta = \int_0^{2\pi}\int_0^{2} r\cdot[(9-r^2)-(r^2+1)] = \int_0^{2\pi}\int_0^{2} -2r^3+8r\,dr\,d\theta = \int_0^{2\pi} -\frac{2r^4}{4}\Big|_0^2 + \frac{8r^2}{2}\Big|_0^2 d\theta = \int_0^{2\pi} 8\,d\theta = 16\pi \text{ u.V.}$$

⑤ $\iiint \sqrt{x^2+y^2+z^2}\,dV$ limitado por $z = \sqrt{x^2+y^2}$; $x^2+y^2+\left(z-\frac{1}{2}\right)^2 = \frac{1}{4}$

$x^2+y^2+z^2 = \rho^2$ Jacob: $\rho^2 \operatorname{sen}\phi$

$x = \rho \operatorname{sen}\phi \cos\theta$ $\quad y = \rho \operatorname{sen}\phi \operatorname{sen}\theta$ $\quad z = \rho \cos\theta$

⑥ $\iiint dA$, delimitado por $x^2+y^2 = 1$, $z = 2$, $z = 1-x^2-y^2$



$$\begin{cases} 0 \le r \le 1 \\ 0 \le \theta \le 2\pi \\ 1-x^2-y^2 \le z \le 2 \end{cases}$$

$\downarrow$

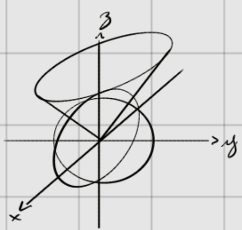
$1-r^2$

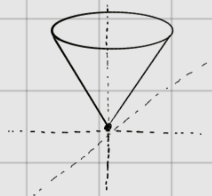
$$\int_0^{2\pi}\int_0^{1}\int_{1-r^2}^{2} r\,dz\,dr\,d\theta = \int_0^{2\pi}\int_0^{1} r+r^3\,dr\,d\theta = \int_0^{2\pi} \frac{3}{4}\,d\theta = \frac{3\pi}{2} \text{ u.v}$$

⑦ O sólido está delimitado por $z = \sqrt{x^2+y^2}$ e abaixo da esfera $x^2+y^2+z^2 = 1$

$x^2+y^2+z^2 = \rho^2$ Jacob: $\rho^2 \operatorname{sen}\phi$

$x = \rho \operatorname{sen}\phi \cos\theta$ $\quad y = \rho \operatorname{sen}\phi \operatorname{sen}\theta$ $\quad z = \rho \cos\theta$

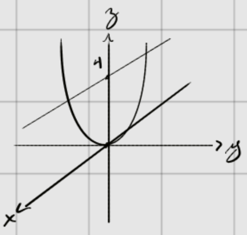




$$\begin{cases} 0 \le \rho \le 1 \\ 0 \le \phi \le \frac{\pi}{4} \\ 0 \le \theta \le 2\pi \end{cases} \qquad \int_0^{2\pi}\int_0^{\frac{\pi}{4}}\int_0^{1} \rho^2 \operatorname{sen}\phi\,d\rho\,d\phi\,d\theta$$

$$\int_0^{2\pi}\int_0^{\frac{\pi}{4}}\int_0^{1} \rho^2 \operatorname{sen}\phi\,d\rho\,d\phi\,d\theta = \int_0^{2\pi}\int_0^{\frac{\pi}{4}} \operatorname{sen}\phi \cdot \frac{\rho^3}{3}\Big|_0^1 d\phi\,d\theta = \int_0^{2\pi}\int_0^{\frac{\pi}{4}} \frac{\operatorname{sen}\phi}{3}\,d\phi\,d\theta = \int_0^{2\pi} \left(1-\frac{\sqrt{2}}{2}\right)\cdot\frac{1}{3}\,d\theta = \left|\frac{1}{3}-\frac{\sqrt{2}}{6}\right|\cdot 2\pi = \frac{2\pi}{3}-\frac{\sqrt{2}\pi}{3} = \frac{\pi}{3}\cdot\left(2-\sqrt{2}\right)$$

⑧ $\iiint (x^2+y^2)^{\frac{1}{2}}$ limitada por $z = x^2+y^2$ e plano $z=4$



$$\begin{cases} 0 \le r \le 2 \\ 0 \le \theta \le 2\pi \\ r^2 \le z \le 4 \end{cases}$$

$$\int_0^{2\pi}\int_0^{2}\int_{r^2}^{4} r^2 \, dz\, dr\, d\theta = \int_0^{2\pi}\int_0^{2} r^2(4-r^2) = \int_0^{2\pi}\int_0^{2} 4r^2 - r^4 \, dr\, d\theta = \int_0^{2\pi} \frac{64}{15}\, d\theta = \frac{128}{15}\pi \,//$$

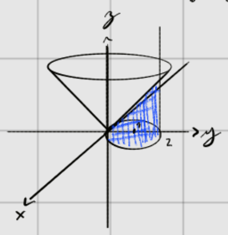
⑨ $$\iiint (x^2+y^2) = \int_0^{\pi}\int_0^{2\operatorname{sen}\theta}\int_0^{r} r^2 \cdot r \, dz\, dr\, d\theta = \int_0^{\pi}\int_0^{2\operatorname{sen}\theta}\int_0^{r} r^3 \, dz\, dr\, d\theta = \int_0^{\pi}\int_0^{2\operatorname{sen}\theta} r^4 \, dr\, d\theta = \int_0^{\pi} (2)^4 \operatorname{sen}^4\theta \, d\theta = 16 \cdot \int_0^{\pi} \operatorname{sen}^4\theta \, d\theta =$$

$x^2+y^2-2y=0$

$x^2+(y-1)^2-1=0$

$x^2+(y-1)^2=1$

↳ Centro é $(0,1)$



$$\begin{cases} 0 \le z \le r \\ 0 \le r \le 2\operatorname{sen}\theta \\ 0 \le \theta \le \pi \end{cases}$$

↳ Esse ângulo devido a localização que o cilindro se encontra